

Annex B

Cover Page for 2009 MJC H2 Prelim Paper 1 & 2

Qn/No	Topic Set	Paper Number	Answers	Marks
1	Method of Difference	9740/01	$\frac{2}{r^2 + 6r + 8} = \frac{1}{r + 2} - \frac{1}{r + 4};$ $\frac{7}{12} - \left(\frac{1}{n+3} + \frac{1}{n+4} \right);$ $n + \frac{70}{3} - 40 \left(\frac{1}{n+3} + \frac{1}{n+4} \right)$	6
2	Mathematical Induction – Recurrence Relation	9740/01		
3	Integration by substitution	9740/01	$\frac{16\sqrt{2}}{3}$	7
4	Differential Equation	9740/01		7
5	Vectors – Exclude Planes and Lines	9740/01	(i) $k = 2; \overrightarrow{OC} = 3\mathbf{b} - 2\mathbf{a}$ (iii) $\theta = \frac{\pi}{3}$ or 60° (iv) $\begin{pmatrix} -2 \\ 2 \\ -2 \end{pmatrix}$	9
6	Complex Numbers – Loci	9740/01	(ii) $z = 1 - \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$ (iii) $\sqrt{2} + 1$ (iv) $-\frac{7\pi}{8}$	9
7	Complex Numbers – Exclude Loci	9740/01	(i) $-15 - 8i$ $z = 2 + 2i, z = -14 - 2i$ (ii) $\Rightarrow z = \frac{1}{2}e^{i\frac{\pi}{3}}, \frac{1}{4}$ or $\frac{1}{2}e^{-i\frac{\pi}{3}}$	10
8	Binomial Expansion and Maclaurin's Series	9740/01	(a) $2^n + n2^{n-1}x + [n(n+1)2^{n-3}]x^2$ (b) (i) $\frac{1}{3}x + \frac{2}{9}x^2 + \frac{4}{27}x^3 + \dots$ (ii) $\left\{ x : x \in \mathbb{R}, -\frac{3}{2} < x < \frac{3}{2} \right\}$ (iii) $n! \left[\frac{1}{3} \left(\frac{2}{3} \right)^{n-1} \right]$	10
9	Vectors – Planes and Lines	9740/01	(i) $\alpha = -1, \beta = 3$	

			(ii) $l_1 : r = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3b-a \\ 3a-b \\ -8 \end{pmatrix}, \lambda \in \mathbb{R}$ (iii) $a = 2, b = -2$ (iv) $p - q = 1$	11
10	AP, GP	9740/01	(a) $-275a$ (b) (i) 1 hour (ii) 5.25 hours (iii) 33 rd lesson	11
11	Differentiation (Applications and Graphs of f')	9740/01	(b) (ii) $\frac{2\sqrt{3}}{3}a^2$ (c) $x = 12.6$ cm	14
	Pure Maths			
1	Integration by parts	9740/02	(i) $-e^{\cos x} \sin x$ (ii) $-2e^{\cos x} \cos x + 2e^{\cos x} + C$	4
2	Graphs	9740/02		9
3	Equations (System of equations) and Inequalities	9740/02	(a) $y = 2\sqrt{x} + \frac{3}{x} + x^2 - 15$ (b) (ii) $x < -2$ or $0 < x < 1$ (ii) $x = 1$ or $0 < x < e^{-1}$ or $x > e$	9
4	Functions	9740/02	(i) Largest $k = 1$ (ii) $f^{-1} : x \mapsto \frac{x - \sqrt{x^2 - 16}}{2}, x \geq 2$ minimum $\lambda = 9$ $R_{hg} = (\ln 2, \ln 11]$	9
5	Integration – Area and Volume	9740/02	(b) $15 + 3 \ln \frac{1}{2}$	9
	Statistics			
6	Sampling Methods	9740/02		4
7	Hypothesis Testing	9740/02	p -value = 0.0687; do not reject H_0	6
8	Permutation and Combination	9740/02	(a) (i) 1680 (ii) 6720 (b) (i) 42336000 (ii) 65318400	8
9	Probability	9740/02	(a) (i) $\frac{2}{19}$ (ii) $\frac{15}{38}$	8

			(b) (i) $\frac{4}{5}$ (ii) $\frac{79}{380}$ (iii) $\frac{79}{95}$	
10	Binomial and Poisson Distributions (include approximation)	9740/02	(i) 0.413 (ii) 0.00141 (iii) 0.0290 or 0.0291 (iv) 0.280	10
11	Correlation and Regression	9740/02	(a) (i) 0.892 (ii) 0.960 (iii) $F = 0.359 + 0.0292x$ $x = 205 + 31.5F$ (iv) 58.4 seconds	12
12	Normal Distribution (include Sample Mean CLT)	9740/02	(a) $\mu = 170$; $\sigma^2 = 58.6$ (i) 0.394 (ii) 0.968 (b) Least $n = 166$	12